OBJECT DETECTION ANF DEPTH ESTIMATION

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1. Introduction

Concept:

Object detection and monocular depth estimation are key tasks in robotics, autonomous driving, and embedded systems.
This project combines YOLOv13 for real-time object detection and TST (Token Sharing Transformer) for accurate depth estimation fro m a single RGB image.
The goal is to achieve high performance on resource-limited devic es like NVIDIA Jetson Nano/TX2.

method:

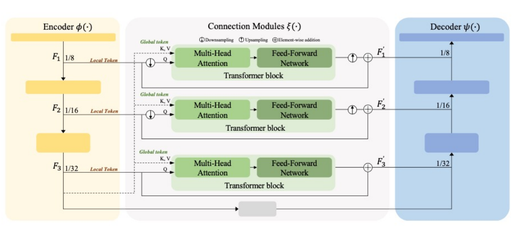
1. input RGB image
2. Detect objects
3. estimate depth
4. Remove outlier
5. Output

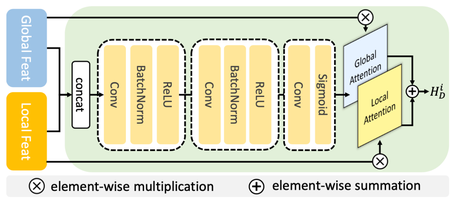
Main concepts in this project:

1. Time Complexity
2. Memory usage

2. Practical Basis

1. Object detector: YOLO13s
2. Depth estimator: TST model





3. Encoder : Mobilenetv2
4. Decoder : SFF + Upsampling
5. Remove outlier
   1. IQR
   2. Z-score
   3. Trimmed
   4. Adaptive

3. implemention

1. Data augmentation (with 0.5 probability)
   1. Random horizontal flips
   2. Random brightness
   3. Contrast
   4. Gamma
   5. Hue
   6. Saturation delta
   7. Vertical cutdepth(0.25 probability)

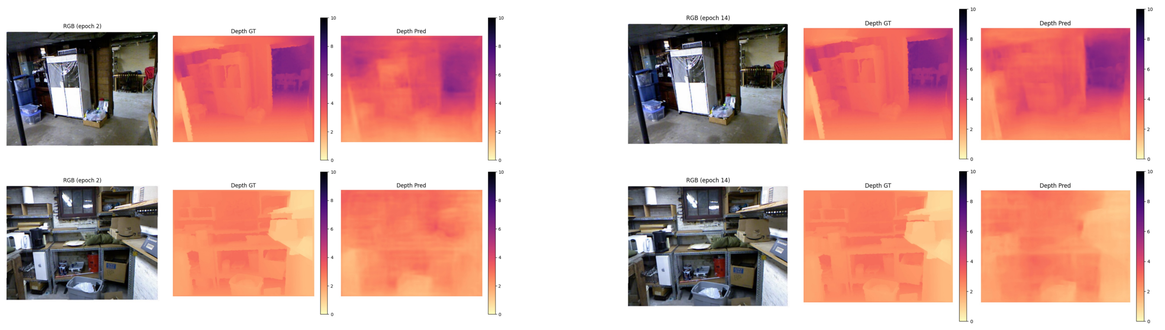
2. Implementation details :
   1. Optimizer : Adam
   2. Depth :0-10 m
   3. Learning rate = 0.0003
   4. Epoch = 14
3. Loss : Scale-invariant log scale loss

$$L = \frac{1}{n}\sum_i \left(d_i^2 - \frac{1}{2n^2}\sum_i d_i^2\right)$$

$$d_i = \log(y_i) - \log(\hat{y}_i)$$

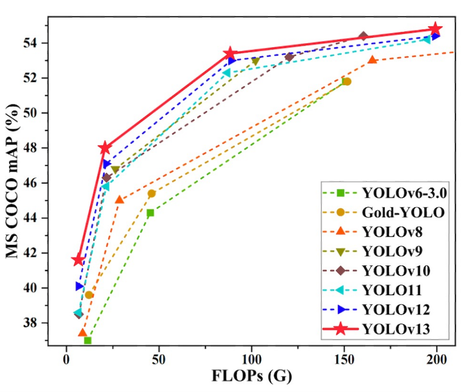
y_i : real pixel of i

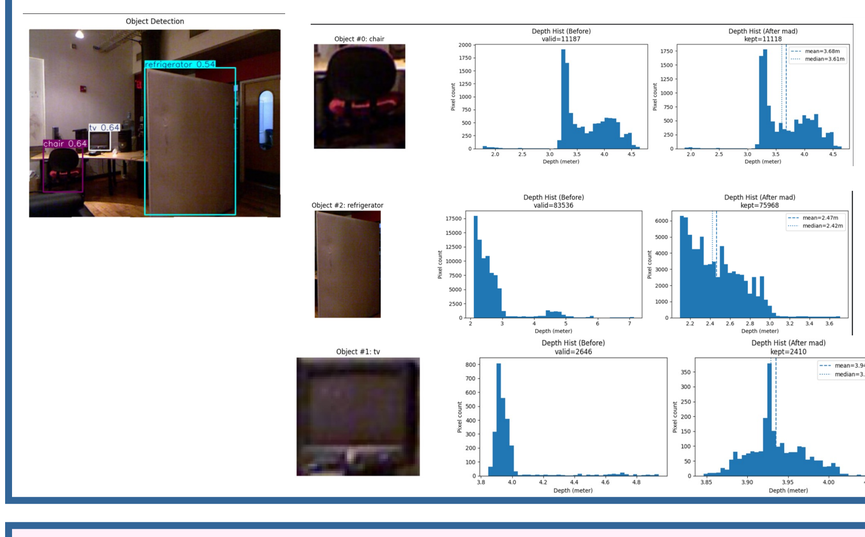
$\hat{y}_i$: predicted pixel of i



4. Results

Method	δ1 ↑	AbsRel ↓	SqRel ↓	RMSE ↓
TuMDE	0.806	0.143	0.096	0.485
FastDepth [5]	0.781	0.157	0.117	0.536
GuidedDepth	0.840	0.129	0.088	0.449
GuidedDepth-S	0.817	0.140	0.095	0.484
Our-depth-model	0.897	0.127	0.088	0.4368





5. Conclusions

The proposed pipeline combines YOLOv13 and TST to achieve accurate object detect ion and monocular depth estimation on embedded devices.
Key advantages:

1. High depth accuracy (AbsRel 0.127) on NYU Depth V2
2. Efficient token-sharing transformer with MobileNetV2 encoder
3. Robust distance extraction using outlier removal
4. Suitable for real-time autonomous systems

Future work: Test on more challenging outdoor datasets, apply knowledge distillatio n for further speedup.